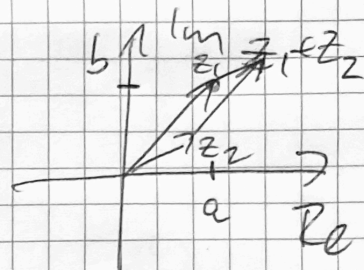


Simple complex functions

- Complex arithmetic

$$z = \underbrace{a}_{\text{Re } z} + i \underbrace{b}_{\text{Im } z} \quad a, b \in \mathbb{R}, \quad i^2 = -1$$

$$z_1 + z_2 = (a_1 + a_2) + i(b_1 + b_2)$$



$$z_1 \cdot z_2 = (a_1 + ib_1)(a_2 + ib_2) =$$

$$= (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + a_2 b_1)$$

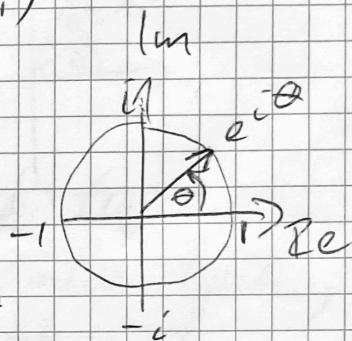
$$|z|^2 = z \cdot \bar{z} = a^2 + b^2$$

$$\frac{1}{i} = -i$$

- Complex exponential

~~$$e^{i\theta} = \cos \theta + i \sin \theta$$~~

$$\theta \in \mathbb{R}$$



General argument e^z

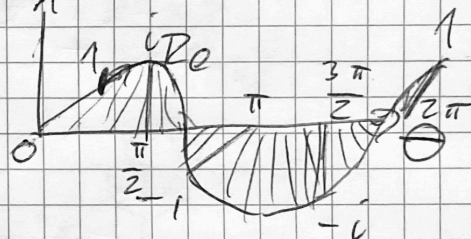
$$e^z = e^{a+ib} = e^a \cdot e^{ib} =$$

$$= e^a \cdot (\cos b + i \sin b)$$

real
part
magnitude

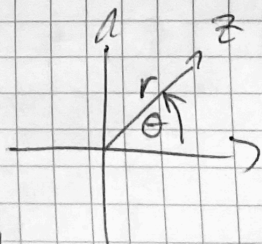
imag
part
phase

of z
of e^z



periodic in $\text{Im } z$

$$e^z = e^{z+2\pi i} = e^z \cdot \underbrace{e^{2\pi i}}_1$$



- Polar form $z = r \cdot e^{i\theta}$

principal argument $-\pi < \theta \leq \pi$

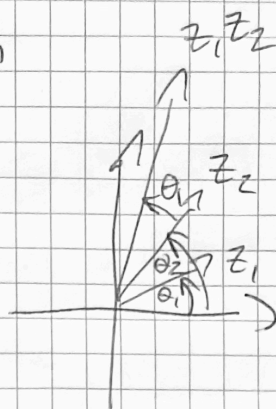
$$r = |z|$$

$$\tan \theta = \frac{b}{a}$$

Multiplication in polar form

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

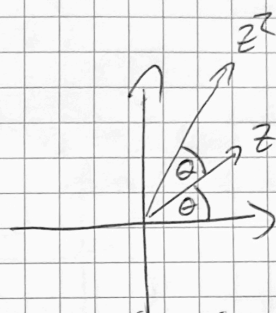
argument added,
magnitude multiplied



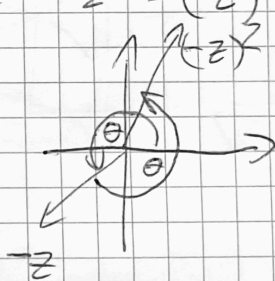
- Example $f(z) = z^2$

$$z = r e^{i\theta}$$

$$z^2 = r^2 e^{2i\theta}$$



Winds around twice
takes every value twice,
because $z^2 = (-z)^2$



- Example $f(z) = \sqrt{z}$

What is $\sqrt{\quad}$?

def: $w = \sqrt{z}$ if $w^2 = z$

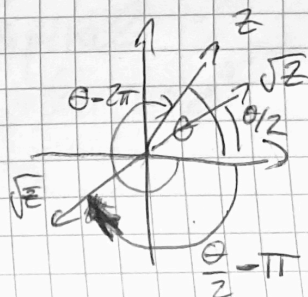
(but we saw that w and $-w$ both work!)

$\text{Arg } z$ $r_w = r_z$, $2\theta_w = \theta_z \pmod{2\pi}$

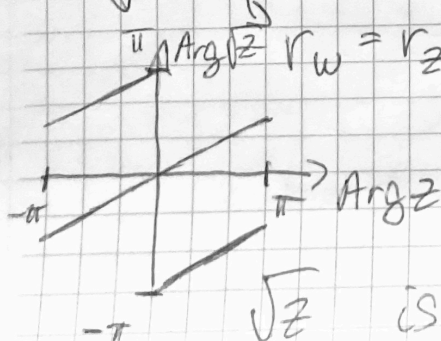
$\theta_w = \frac{\theta_z}{2} \pmod{\pi}$

$\theta_w = \frac{\theta_z}{2} \text{ or } \frac{\theta_z}{2} + \pi$

$\sqrt{z}$ is double valued



rotate graph



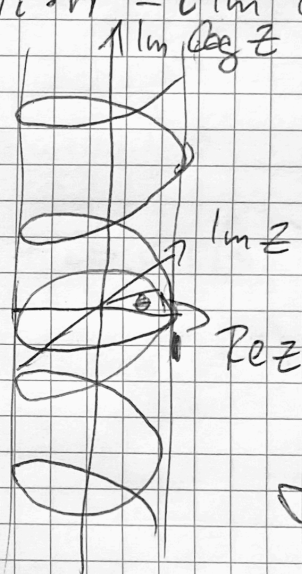
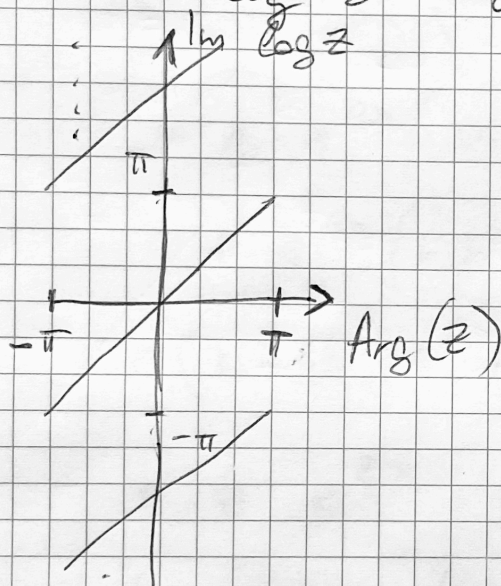
• Example complex logarithm

$$\log z = w \quad \text{if} \quad \exp w = z$$

If true for w , true for $w + 2\pi i \cdot n \quad n \in \mathbb{Z}$
infinite-valued!

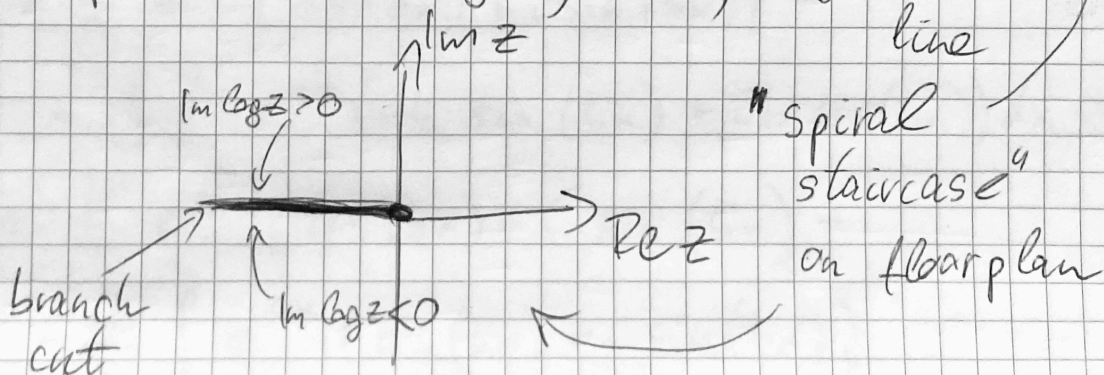
Let $z = \cancel{r} e^{i\theta} = \text{Arg}(z)$

$$\log z = i\theta + 2\pi i \cdot n = i \text{Im} \log z$$



Make it unique: $-\pi < \text{Im} \log z \leq \pi$
principal log

Sample where $\text{Arg}(z) = \pm\pi$, negative real line



• Prove that $\int e^{zx} dx = \frac{1}{z} e^{zx}$
using the real and imag. part of e^{zx}

1) Using derivative

equivalent to $\frac{d}{dx} \left(\frac{1}{z} e^{zx} \right) = e^{zx}$

$$\begin{aligned} \frac{d}{dx} e^{zx} &= \frac{d}{dx} \left(e^{ax} (\cos(bx) + i \sin(bx)) \right) = \\ &= a e^{ax} (\cos(bx) + i \sin(bx)) + \\ &\quad + e^{ax} (-b \sin(bx) + ib \cos(bx)) = \\ &= (a + ib) e^{ax} (\cos(bx) + i \sin(bx)) = \\ &= z e^{zx} \quad \checkmark \end{aligned}$$

2) Integration by parts

$$\begin{aligned} \int \underbrace{e^{ax}}_u \underbrace{(\cos(bx) + i \sin(bx))}_v dx &= \\ &= \frac{1}{a} e^{ax} (\cos(bx) + i \sin(bx)) - \\ &\quad - \int \frac{1}{a} e^{ax} (-b \sin(bx) + ib \cos(bx)) dx = \\ &= \frac{1}{a} e^{ax} (\cos(bx) + i \sin(bx)) - \\ &\quad - \frac{ib}{a} \int e^{ax} (\cos(bx) + i \sin(bx)) \end{aligned}$$

rearrange the equation,
the same integral appears on both
sides

$$\left(1 + \frac{ib}{a}\right) \int e^{ax} (\cos bx + i \sin bx) dx = \frac{1}{a} e^{ax} (\cos bx + i \sin bx)$$

$$\int e^{ax} (\cos(bx) + i \sin(bx)) dx =$$

$$= \frac{1}{a + ib} e^{ax} (\cos(bx) + i \sin(bx))$$

$$\int e^{zx} dx = \frac{1}{z} e^{zx} \quad \square$$

Some Fourier transforms

1.)

$$f(x) = \begin{cases} 1 & -b < x < c \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x) e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_b^c dx e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \left[\frac{1}{-i\omega} e^{-i\omega x} \right]_{x=b}^c = \\ &= \frac{i}{\sqrt{2\pi}\omega} (e^{-i\omega c} - e^{-i\omega b}) \end{aligned}$$

$$\text{If } b = -c \text{ then } \hat{f}(\omega) = \frac{i}{\sqrt{2\pi}\omega} (e^{-i\omega c} - e^{i\omega c}) = \sqrt{\frac{2}{\pi}} \frac{\sin(\omega c)}{\omega}$$

2.)

$$f(x) = \begin{cases} e^{-ax} & (a > 0) \quad x > 0 \\ 0 & \text{otherwise} \end{cases} \quad (*)$$

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x) e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dx e^{-ax} e^{-i\omega x} \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-(a+i\omega)x}}{-(a+i\omega)} \right]_{x=0}^{\infty} = \frac{1}{\sqrt{2\pi}} \left(0 - \frac{1}{-(a+i\omega)} \right) = \frac{1}{\sqrt{2\pi}} \frac{1}{a+i\omega} \end{aligned}$$

3.)

$$g(x) = e^{-a|x|} \quad a > 0$$

Note: $g(x) = f(x) + f(-x)$ where $f(x)$ is given in (*)

$$\begin{aligned} \text{Then } \hat{g}(\omega) &= \hat{f}(\omega) + \hat{f}(-\omega) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{a+i\omega} + \frac{1}{a-i\omega} \right) \\ &= \frac{1}{\sqrt{2\pi}} \frac{2a}{a^2 + \omega^2} \end{aligned}$$

or:

$$f(x) = e^{-a|x|} \quad a > 0$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-a|x|} e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 dx e^{+ax} e^{-i\omega x} + \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dx e^{-ax} e^{-i\omega x}$$

$$\begin{aligned} & y = -x \\ & = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dy e^{-ay} e^{i\omega y} + \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dx e^{-ax} e^{-i\omega x} \end{aligned}$$

$$\text{Here } \int_0^{\infty} dx e^{-ax} e^{-i\omega x} = \left[\frac{e^{-ax-i\omega x}}{-a-i\omega} \right]_{x=0}^{\infty} = 0 - \frac{1}{-a-i\omega} = \frac{1}{a+i\omega}$$

1st term: $\omega \rightarrow -\omega$, so

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{a-i\omega} + \frac{1}{a+i\omega} \right) = \frac{1}{\sqrt{2\pi}} \frac{2a}{\omega^2 + a^2}$$

Reverse direction:

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \hat{f}(\omega) e^{+i\omega x}$$

So,

$$e^{-a|x|} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{1}{\sqrt{2\pi}} \frac{2a}{\omega^2 + a^2} e^{i\omega x} \stackrel{\omega \rightarrow -\omega}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{1}{\sqrt{2\pi}} \frac{2a}{\omega^2 + a^2} e^{-i\omega x}$$

swapping ω and x :

$$\mathcal{F}\left(\frac{1}{x^2 + a^2}\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \frac{e^{-i\omega x}}{x^2 + a^2} = \sqrt{\frac{\pi}{2}} \frac{e^{-a|\omega|}}{a}$$

We don't need to compute this integral!

$$f(t) = e^{i\omega_0 t}$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt e^{i\omega_0 t} e^{-i\omega t} = \sqrt{2\pi} \delta(\omega - \omega_0)$$

Reverse:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \sqrt{2\pi} \delta(\omega - \omega_0) e^{+i\omega t} = e^{i\omega_0 t} \quad \checkmark$$

$$\mathcal{F}(e^{i\omega_0 t}) = \sqrt{2\pi} \delta(\omega - \omega_0)$$

$$f(t) = \cos(\omega_0 t) = \frac{1}{2} (e^{i\omega_0 t} + e^{-i\omega_0 t})$$

$$\mathcal{F}(\cos \omega_0 t) = \sqrt{\frac{\pi}{2}} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right)$$

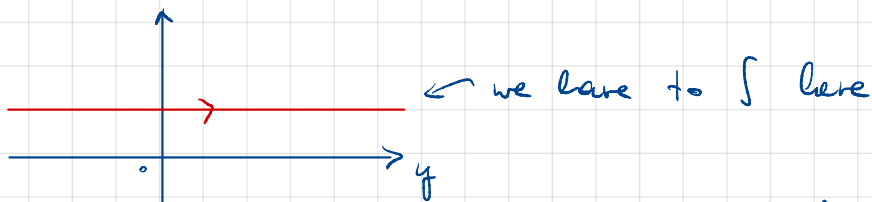
4.) $f(x) = e^{-ax^2}$ ($a > 0$) Gaussian (bell curve)

$$\sqrt{2\pi} \hat{f}(w) = \int_{-\infty}^{\infty} dx e^{-ax^2} e^{-iwx} = \int_{-\infty}^{\infty} dx e^{-a\left(x + \frac{iw}{2a}\right)^2 + \left(\frac{w}{2a}\right)^2}$$

$$= e^{-\frac{w^2}{4a}} \int_{-\infty}^{\infty} dx e^{-a\left(x + \frac{iw}{2a}\right)^2}$$

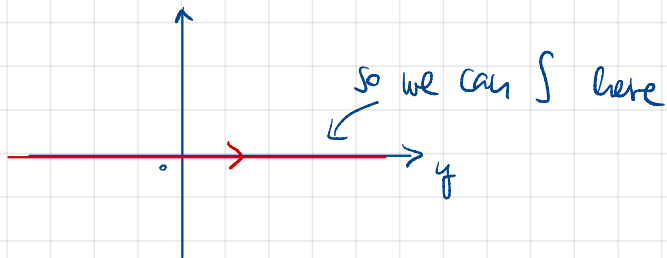
Naïve change of variables: $y = x + \frac{iw}{2a}$

$$= e^{-\frac{w^2}{4a}} \int_{-\infty + \frac{iw}{2a}}^{\infty + \frac{iw}{2a}} dy e^{-ay^2}$$



|| (equals)

Reason: if the integrand has no poles (divergences) and no analytic points the integration contour can be shifted



Now we use

$$\int_{-\infty}^{\infty} dy e^{-ay^2} = \sqrt{\frac{\pi}{a}} \quad (\text{useful to know by heart!})$$

Proof: Consider the square:

$$\int_{-\infty}^{\infty} dy e^{-ay^2} \cdot \int_{-\infty}^{\infty} dz e^{-az^2} = \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz e^{-a(y^2+z^2)}$$

Integral on y - z plane, integrand depends on $r^2 = y^2 + z^2$ only.

Use radial coordinates:

$$= 2\pi \int_0^{\infty} dr r e^{-ar^2} = 2\pi \left[-\frac{1}{2a} e^{-ar^2} \right]_{r=0}^{\infty} = 2\pi \left(0 - \left(-\frac{1}{2a}\right) \right) = \frac{\pi}{a}$$

So,

$$F(e^{-ax^2}) = \hat{f}(\omega) = \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2\pi}} \sqrt{\frac{\pi}{a}} = \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}}$$

5.) $f(x) = e^{-a(x-b)^2}$ $a > 0$

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-a(x-b)^2} e^{-i\omega x} \stackrel{z=x-b}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-az^2} e^{-i\omega(z+b)} = \\ &= \frac{e^{-i\omega b}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-az^2} e^{-i\omega z} = e^{-i\omega b} \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}} \end{aligned}$$

[General rule: $g(x) = f(x-b) \Rightarrow \hat{g}(\omega) = e^{-i\omega b} \hat{f}(\omega)$]

Reverse direction:

$$e^{-a(x-b)^2} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}} e^{-i\omega b} \cdot e^{+i\omega x} \stackrel{\omega=-\omega}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}} e^{+i\omega b} e^{-i\omega x}$$

$\begin{matrix} \omega \rightarrow x \\ x \rightarrow \omega \end{matrix} \rightarrow F\left(e^{-\frac{x^2}{4a}} e^{+ibx}\right) = \sqrt{2a} e^{-a(\omega-b)^2}$

Now directly:

6.) $h(x) = e^{-\alpha x^2 + ibx}$

$$\hat{h}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-\alpha x^2 + ibx} e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-\alpha x^2} e^{-i(\omega-b)x}$$

from Problem 4

$$= \hat{f}(\omega-b) = \frac{e^{-\frac{(\omega-b)^2}{4\alpha}}}{\sqrt{2\alpha}}$$

With $\alpha = 1/4a$ we recover the result in 5).